



WOXSEN UNIVERSITY

School of Technology

PROJECT / RESEARCH REPORT

AI-ENABLED EMERGENCY SAFETY HELMET FOR PROACTIVE WORKER HAZARD DETECTION AND LOCATION-AWARE RISK INTELLIGENCE

Submitted in partial fulfillment of the requirements for the degree of
B. Tech. in Artificial Intelligence and Machine Learning

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CERTIFICATE

This is to certify that the project/research report entitled “AI-Enabled Emergency Safety Helmet for Proactive Worker Hazard Detection and Location-Aware Risk Intelligence” submitted by [Student Name(s)] ([UGID(s)]) in partial fulfillment of the requirements for the award of the degree of B. Tech. in Artificial Intelligence and Machine Learning from Woxsen University is a bona fide record of academic work carried out by the students under my supervision and guidance. The work presented in this report has been prepared for academic purposes and has not been submitted elsewhere for the award of any other degree or diploma.

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DECLARATION OF THE CANDIDATE

We hereby declare that the project/research work entitled “AI-Enabled Emergency Safety Helmet for Proactive Worker Hazard Detection and Location-Aware Risk Intelligence” submitted to the School of Technology, Woxsen University, in partial fulfillment of the requirements for the award of the degree of B. Tech. in Artificial Intelligence and Machine Learning is our original academic work carried out under the guidance of the faculty mentor. We further declare that the technical discussion, system architecture, and proposed methodology presented in this report have been prepared for this project and that no experimental performance values have been fabricated. Where prior research has been used, it is acknowledged through appropriate references.

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ABSTRACT

This research proposes an AI-enabled emergency safety helmet for workers operating in hazardous environments, with underground mining used as the primary use case. The proposed helmet combines worker identification, location/zone tracking, impact and fall sensing, a manual emergency button, environmental hazard sensing, local alarms, and long-range communication with a central monitoring server. Each helmet is assigned a unique identification number so that every event can be associated with a particular worker and operational zone.

The central research contribution is a proactive hazard-intelligence layer. Instead of treating each emergency as an isolated event, the system stores historical incidents together with their location, event type, severity, time, and worker/helmet identifier. If repeated falls, high-impact events, or emergency-button activations occur in the same zone, the AI module can increase that zone's risk score. When another worker enters the affected zone, the system can issue a preventive warning through the helmet and/or control dashboard. This changes the system from a reactive emergency-alert device into a continuously learning safety-support system.

The proposed prototype uses an ESP32 as the controller, an MPU6050 accelerometer/gyroscope for motion analysis, a piezoelectric impact sensor, a prototype MQ-135 gas sensor, a LoRa SX1278 communication module, an emergency push button, LED status indicators, a buzzer, a high-power headlamp, and a rechargeable 3.7 V 18650 battery with TP4056 charging circuitry. The research defines the architecture, sensing logic, communication flow, risk-scoring approach, dashboard requirements, validation plan, and limitations. The work is presented as a prototype/research design; deployment in certified mining environments would require industrial-grade sensors, intrinsic-safety considerations, calibrated gas detection, and field validation.

Keywords: Smart Helmet, Worker Safety, Internet of Things, Artificial Intelligence, Fall Detection, Impact Detection, LoRa, Hazard Mapping, Risk Prediction, Underground Mining, Edge Computing

TABLE OF CONTENTS

1. INTRODUCTION

- 1.1 Background
- 1.2 Motivation
- 1.3 Problem Statement
- 1.4 Research Questions
- 1.5 Objectives
- 1.6 Scope
- 1.7 Limitations
- 1.8 Organization of the Report

2. LITERATURE REVIEW AND TECHNOLOGY BACKGROUND 2.1 Smart Helmets and IoT-Based Worker Safety

- 2.2 Wearable Fall and Impact Detection
- 2.3 Gas and Environmental Hazard Monitoring
- 2.4 LoRa Communication in Underground Environments
- 2.5 Location Tracking in Mines
- 2.6 Artificial Intelligence for Proactive Risk Intelligence
- 2.7 Research Gap
- 2.8 Proposed Contribution

3. PROPOSED SYSTEM AND METHODOLOGY

- 3.1 System Overview
- 3.2 Hardware Architecture
- 3.3 Helmet Identification and Worker Association
- 3.4 Sensing and Event Detection
- 3.5 Fall Detection Logic
- 3.6 Impact Detection Logic
- 3.7 Emergency Button
- 3.8 Environmental Hazard Detection
- 3.9 Location/Zone Tracking
- 3.10 LoRa Communication
- 3.11 Server and Dashboard
- 3.12 AI-Based Hazard Hotspot Intelligence
- 3.13 Risk-Score Formulation
- 3.14 End-to-End Workflow
- 3.15 Data Packet Structure
- 3.16 Alert Prioritization

4. RESULTS, VALIDATION PLAN AND DISCUSSION

- 4.1 Prototype Output
- 4.2 Validation Strategy
- 4.3 Test Cases

4.4 Expected System Behaviour

4.5 AI Hazard-Hotspot Scenario

4.6 Error and False-Alarm Handling

4.7 Discussion

5. CONCLUSION AND FUTURE WORK

5.1 Summary of Contributions

5.2 Conclusion

5.3 Limitations

5.4 Future Work

REFERENCES

APPENDIX A – COMPONENTS

APPENDIX B – SAMPLE EVENT LOG

APPENDIX C – PRESENTATION POINTS

LIST OF TABLES

Table 1.1: Research Objectives

Table 2.1: Comparison of Related Safety Systems

Table 3.1: Proposed Hardware Components

Table 3.2: Helmet-to-Server Event Packet

Table 3.3: Alert Priority Logic

Table 4.1: Validation Test Cases

Table 4.2: Expected Prototype Observations

Table 5.1: Limitations and Future Improvements

LIST OF FIGURES

Figure 1.1: Conceptual Smart Safety Helmet

Figure 3.1: Proposed System Architecture

Figure 3.2: Helmet Internal/Exploded Component View

Figure 3.3: Control Room Dashboard and Alert Flow

Figure 3.4: AI Hazard-Hotspot Learning Cycle

Figure 4.1: Prototype Output and Dashboard Scenarios

1. INTRODUCTION

1.1 Background

Workers in mining, construction, manufacturing, tunnelling, and other industrial environments can be exposed to multiple hazards simultaneously. Physical impacts, falls, hazardous gases, poor visibility, equipment-related incidents, and delayed emergency communication can increase the severity of an accident. Conventional safety helmets provide essential passive protection, but a conventional helmet generally does not know whether a worker has fallen, where the worker is located, whether the worker has manually requested assistance, or whether a particular work zone has recently experienced repeated incidents.

Recent research demonstrates that IoT-enabled helmets can integrate sensing, communication, environmental monitoring, and location tracking. For example, published work has proposed smart helmets for mine workers that combine sensors with wireless communication and control-room monitoring. [1], [2] The proposed project extends this direction by adding a cross-worker learning mechanism that converts repeated incident history into location-specific preventive warnings.



Figure 1.1: Conceptual smart safety helmet used as the visual design reference.

1.2 Motivation

- Reactive safety systems normally respond after an event. A worker falls, presses an emergency button, or a hazard is detected, and an alert is generated.
- The proposed system aims to add a preventive layer. If multiple incidents occur in the same physical zone, the system should learn that the zone is associated with elevated risk.
- Workers should not need to manually communicate their exact identity during an emergency. A unique helmet ID allows the server to associate sensor data with the assigned worker.
- Underground environments can have unreliable conventional wireless connectivity. Long-range low-power communication such as LoRa is therefore investigated as a communication layer.
- A control-room dashboard can provide a unified view of worker status, helmet status, event history, location/zone, battery state, and active alerts.

1.3 Problem Statement

The problem addressed in this research is the design of a connected worker-safety helmet capable of detecting emergency events, identifying the affected worker, communicating the event to a monitoring server, and learning from historical incidents to identify recurring hazardous zones. Existing systems may provide one or more of these capabilities, but the proposed research focuses on integrating immediate emergency response with location-aware, cross-worker hazard intelligence.

A critical design issue is that the helmet cannot directly “see” an injury. Sensors observe measurable proxies such as acceleration, orientation change, impact, inactivity, gas concentration, and manual emergency-button activation. The AI layer therefore estimates an emergency or risk condition from sensor evidence rather than claiming medical diagnosis.

1.4 Research Questions

- 1.RQ1: Can a wearable helmet architecture combine impact, motion, environmental, and manual emergency sensing into a unified worker-safety event?
- 2.RQ2: Can unique helmet IDs and zone-level communication provide useful real-time worker/event visibility to a control room?
- 3.RQ3: Can repeated incidents from multiple helmets be aggregated to identify hazardous work zones?
- 4.RQ4: Can a risk score derived from historical incidents generate preventive warnings for workers entering high-risk zones?
- 5.RQ5: What technical limitations must be addressed before deployment in real underground industrial environments?

1.5 Objectives

No.	Objective
1	Design a connected safety helmet architecture for hazardous workers.
2	Assign every helmet a unique ID and associate it with a worker record.
3	Detect abnormal motion, suspected falls, and strong

	impacts using inertial and impact sensors.
4	Provide a manual emergency button for immediate worker-initiated assistance.
5	Monitor environmental conditions using a prototype gas-sensing subsystem.
6	Transmit event data to a control server using LoRa-based communication.
7	Represent workers and incidents on a dashboard using zone/location information.
8	Aggregate historical incidents and generate location-specific risk scores.
9	Warn subsequent workers when they approach a historically risky zone.
10	Define a validation methodology without fabricating unmeasured performance.

1.6 Scope

The primary use case is underground coal mining, but the architecture can be adapted to construction sites, tunnels, factories, warehouses, and other hazardous workplaces. The scope includes the helmet electronics, local event detection, communication, server-side event storage, dashboard visualization, risk scoring, and preventive warning logic. The project does not claim certification for actual mine deployment.

1.7 Limitations

- ☐ The prototype gas sensor shown in the concept is a general-purpose MQ-135 sensor. Mine deployment would require calibrated, application-specific and safety-certified gas sensors.
- ☐ LoRa provides communication but does not inherently provide absolute underground coordinates. Zone-level location requires gateways, anchors, beacons, RSSI-based estimation, or another positioning layer.
- ☐ Fall detection may generate false positives from activities such as jumping, dropping the helmet, climbing, or sudden movement.
- ☐ AI risk scores depend on sufficient historical data. A new site with little incident history cannot immediately provide strong predictive warnings.
- ☐ The conceptual prototype is not a substitute for statutory PPE, mine safety procedures, rescue protocols, or certified emergency communication systems.

1.8 Organization of the Report

Chapter 2 reviews related research and the technologies supporting the proposed design. Chapter 3 presents the hardware, communication architecture, sensing logic, server workflow, and AI hazard-hotspot model. Chapter 4 presents the prototype outputs, validation plan, expected behaviour, and discussion. Chapter 5 concludes the study and identifies future work.

2. LITERATURE REVIEW AND TECHNOLOGY BACKGROUND

2.1 Smart Helmets and IoT-Based Worker Safety

Smart helmets have been investigated as wearable IoT platforms that combine traditional PPE with sensing and communication. Singh et al. presented an IoT-enabled helmet for mine workers incorporating environmental sensing and worker location monitoring, demonstrating the feasibility of integrating multiple safety functions into a helmet. [1] Another IEEE study described an IoT-based smart helmet for automated multi-parametric monitoring of underground miners using wireless connectivity to a control room. [2] These studies establish the feasibility of connected PPE, but the proposed research emphasizes a further layer: learning from incident patterns across multiple workers.

2.2 Wearable Fall and Impact Detection

Fall detection commonly uses accelerometers and gyroscopes because a fall can produce a characteristic sequence involving sudden acceleration, orientation change, and a subsequent period of low movement. Research using wearable inertial sensors has demonstrated that machine-learning classifiers can distinguish falls from ordinary activities, although real-world performance depends strongly on sensor position, data quality, and the diversity of activities represented in the training data. [3], [4]

For the proposed helmet, the MPU6050 provides three-axis acceleration and three-axis angular velocity. A piezoelectric impact sensor can provide an additional signal for sharp mechanical shocks. Combining these signals is preferable to relying on a single threshold. The first prototype can use threshold-based event logic, while a later version can train a lightweight classifier on labeled helmet motion sequences.

2.3 Gas and Environmental Hazard Monitoring

Environmental monitoring is particularly relevant in underground mines because harmful gases and changing atmospheric conditions can create serious hazards. Research on IoT-enabled mining helmets has explored real-time environmental sensing and warning mechanisms. [1] The proposed prototype includes an MQ-135 sensor as a broad air-quality/hazard prototype. However, an MQ-135 should not be treated as a calibrated methane or carbon-monoxide safety instrument. A production system would require sensors selected and calibrated specifically for the gases of interest, with appropriate certification and alarm thresholds.

2.4 LoRa Communication in Underground Environments

LoRa is attractive for low-power, long-range telemetry, but underground propagation is strongly affected by tunnel geometry, materials, turns, obstacles, antenna placement, and network topology. Research has specifically evaluated LoRa for underground coal mines and reported useful communication behaviour while also demonstrating that repeaters or boosters may be required in some layouts. [5] Other work has investigated LoRa-based location networks for

underground mining personnel and equipment. [6] Measurements in underground gold mines have also shown that LoRa can propagate effectively in certain tunnel conditions, while still requiring site-specific modelling. [7]

Therefore, this project uses LoRa as the proposed communication layer rather than assuming that a single gateway will cover an entire mine. A practical deployment can use strategically placed gateways/repeaters or a linear network architecture.

2.5 Location Tracking in Mines

Location is essential to rescue response and risk mapping. GPS may not provide reliable positioning underground. The proposed system therefore represents location primarily as a zone, tunnel, section, or gateway-defined area. A helmet can transmit its ID and communication information; the infrastructure can infer the worker's current zone using gateway association, RSSI, anchors, beacons, or a dedicated indoor-positioning technology. The dashboard can then display "Section B – Tunnel 4" rather than claiming unsupported latitude/longitude.

2.6 Artificial Intelligence for Proactive Risk Intelligence

The distinguishing AI feature of the proposed system is not simply classifying one fall event. The system accumulates events from a population of helmets. Suppose Helmet H-021 reports a fall in Zone B, followed by two emergency-button activations from other workers in the same zone and another impact event within a defined time period. The server can increase the risk score of Zone B. Repeated incidents, severity, recency, and environmental warnings can contribute to the score.

This creates a feedback loop: detect event → record event → aggregate events by zone → estimate zone risk → warn future workers → observe subsequent outcomes → update the risk estimate. The design therefore moves toward predictive/preventive safety rather than only post-incident notification.

2.7 Research Gap

Area	Typical existing approach	Gap addressed by proposed system
Worker monitoring	Individual sensor/helmet status	Unified multi-helmet identity and event history
Emergency alerts	Immediate alert after a fall or button press	Immediate response plus historical risk intelligence
Location	GPS or fixed location monitoring	Zone-level location suitable for underground environments
AI	Event classification or health monitoring	Cross-worker spatial-temporal hazard learning
Warnings	Reactive alarm	Preventive warning to later workers entering risky zones
Dashboard	Current status	Current status + historical hotspot/risk view

2.8 Proposed Contribution

The main contribution is a system-level architecture that treats each helmet as both a safety device and a data source for collective hazard intelligence. The helmet detects and reports events, while the server learns where incidents repeatedly occur. This creates a shared safety memory for the workplace: one worker's incident can become a warning for the next worker.

3. PROPOSED SYSTEM AND METHODOLOGY

3.1 System Overview

The proposed system consists of four layers: (1) wearable helmet hardware, (2) local edge processing, (3) wireless communication and location infrastructure, and (4) server-side monitoring and AI analytics. Each helmet continuously samples selected sensors, detects local events, assigns the helmet ID and zone information, and transmits an event packet. The server stores the event and updates the worker/zone state. The AI layer aggregates historical data to identify recurring risk patterns.



Figure 3.1: Proposed control-room dashboard, alert scenarios and end-to-end system flow.

3.2 Hardware Architecture

Component	Role in the proposed system	Important consideration
ESP32 development board	Main microcontroller; reads sensors, applies event logic, controls alarms and communication	Prototype controller; production design requires industrial qualification
MPU6050	3-axis accelerometer + 3-axis gyroscope for motion/fall analysis	Requires calibration and mounting stability
Piezoelectric impact sensor	Detects sharp mechanical shock/impact	Needs mechanical characterization to reduce false triggers
MQ-135 prototype gas sensor	Prototype air-quality/hazard sensing	Not a certified methane/CO detector; requires replacement/calibration for deployment
LoRa SX1278	Long-range low-power telemetry	Requires site-specific RF planning

		and gateway/repeater infrastructure
Emergency push button	Manual worker emergency request	Should be physically accessible and protected from accidental activation
LED indicator	Local status indication: normal/warning/emergency	Use clear and distinguishable states
Piezo buzzer	Local audible warning	Alarm level should be safe and standardized
LED headlamp	Worker visibility in dark areas	Should have suitable brightness and power management
18650 3.7 V battery	Portable power source	Needs protected cell, thermal and charging safeguards
TP4056 charging module	Prototype charging circuit	Production charging/protection must meet electrical safety requirements



Figure 3.2: Conceptual internal/exploded component arrangement of the helmet.

3.3 Helmet Identification and Worker Association

Every helmet receives a unique identifier such as SSH-001, SSH-002, or CM-021. The server maintains a worker registry mapping Helmet ID → Worker ID → Assigned zone/shift → status. When a worker changes helmets, the assignment can be updated without changing the hardware

identity. Every event packet includes the Helmet ID, preventing ambiguity when several workers trigger alarms simultaneously.

3.4 Sensing and Event Detection

The helmet samples motion and environmental signals at a suitable frequency and maintains a short rolling window of sensor data. Local processing looks for patterns associated with impact, fall, gas warning, or manual emergency. Local processing is important because the helmet should still activate its LED/buzzer even if communication to the server is temporarily unavailable.

3.5 Fall Detection Logic

A first prototype can use a multi-stage rule: (1) detect acceleration magnitude above an impact threshold, (2) check for a significant orientation change using gyroscope data, and (3) check for reduced movement for a short post-event interval. If multiple conditions are satisfied, the helmet marks the event as a suspected fall. A later AI classifier can replace or supplement these rules using labeled time-series data.

A simplified acceleration magnitude is calculated as: $a = \sqrt{a_x^2 + a_y^2 + a_z^2}$. The system can compare a to a calibrated baseline and use changes in angular velocity to distinguish a real fall-like sequence from routine movement. Thresholds must be established experimentally for the actual helmet, worker activities, and sensor mounting position.

3.6 Impact Detection Logic

The piezoelectric impact sensor provides a second signal for sudden mechanical shocks. An impact event can be declared when the piezo signal exceeds a calibrated threshold and the IMU simultaneously reports a high acceleration or abrupt orientation change. This sensor fusion approach is intended to reduce false positives caused by vibration or ordinary motion.

3.7 Emergency Button

The emergency button is a direct human override. When pressed, the helmet immediately activates the emergency state and sends an emergency packet containing the Helmet ID, worker identity, zone, timestamp, and event type. The manual emergency channel is important because not every dangerous situation produces a measurable fall or impact.

3.8 Environmental Hazard Detection

The prototype gas subsystem continuously samples air-quality data and compares it with calibrated warning levels. A warning state can trigger a yellow LED and intermittent buzzer, while a critical state can trigger a red emergency state and server alert. In a production mining system, separate calibrated sensors should be used for methane, carbon monoxide, oxygen deficiency, and other gases relevant to the site. The AI system should also distinguish sensor faults from genuine environmental changes.

3.9 Location/Zone Tracking

The server needs a location label for every event. In underground environments, the recommended first implementation is zone-level localization. Each LoRa gateway/repeater is mapped to a known mine section or tunnel. When a helmet communicates through a particular infrastructure node, the server assigns the corresponding zone. More advanced versions can use multiple gateways and RSSI/SNR or dedicated anchors to estimate a finer location.

3.10 LoRa Communication

The helmet transmits compact telemetry packets rather than continuous high-bandwidth data. A typical emergency packet contains Helmet ID, event type, severity, zone ID, timestamp, battery percentage, sensor summary, and sequence number. LoRa is suitable for such small packets, but coverage must be tested on the actual site. Underground studies show that propagation can vary significantly with tunnel geometry and non-line-of-sight conditions. [5]–[7]

3.11 Server and Dashboard

The server maintains four primary data views: live workers, active alerts, historical events, and zone risk. The dashboard shown in the concept includes a worker ID, event type, impact value, gas level, battery state, signal strength, time, and zone. A map panel visualizes the affected tunnel/section and highlights active alerts.

3.12 AI-Based Hazard Hotspot Intelligence

The proposed AI module operates on the historical event database. Each event is represented as a record with zone, time, type, severity, and outcome. The system aggregates these records into rolling zone-level features such as number of impacts, number of suspected falls, emergency-button activations, gas warnings, and recency of incidents. A lightweight risk model then estimates whether a zone should be classified as normal, caution, high-risk, or critical.

For a research prototype, the AI can begin with a transparent weighted risk model and later be replaced with machine learning such as logistic regression, gradient boosting, random forest, or a time-aware model. A transparent first model is preferable because safety warnings need explainable reasons.

3.13 Risk-Score Formulation

Let $R_z(t)$ be the risk score of zone z at time t . A prototype formulation is:

$$R_z(t) = wI \cdot I + wF \cdot F + wE \cdot E + wG \cdot G + wD \cdot D$$

where I represents normalized impact-event frequency/severity, F represents fall events, E represents emergency-button activations, G represents environmental hazard severity, and D represents a recency/decay factor. The weights wI , wF , wE , wG , and wD are calibrated using historical validation data. The score can be bounded to 0–100 for dashboard display.

A key safety principle is that a single serious event should not be ignored simply because the historical score is low. Immediate emergency alerts remain event-driven. The risk score is an additional preventive layer and must never suppress an emergency event.

3.14 End-to-End Workflow

6. Worker wears a helmet with a unique Helmet ID.
7. Helmet samples IMU, impact, gas, and button inputs.
8. ESP32 performs local preprocessing and event detection.
9. If an emergency is detected, the local LED/buzzer activates immediately.
10. The ESP32 creates a compact event packet containing ID, event type, severity, time, and zone information.
11. LoRa transmits the packet to the appropriate gateway/repeater.
12. The server authenticates the helmet ID, stores the event, updates worker status, and displays the alert.
13. The AI risk engine aggregates the event with historical events from other helmets.
14. If the zone risk increases above a configured threshold, the zone is marked as a hazard hotspot.
15. When another worker enters the hotspot, a preventive warning is sent to the helmet and shown on the dashboard.
16. New outcomes are stored and used to update future risk estimates.

3.15 Data Packet Structure

Field	Example	Purpose
Helmet_ID	SSH-021	Uniquedevic identity
Worker_ID	WK-021	Assigned worker identity
Zone_ID	SEC-B-T04	Location/zone label
Timestamp	2026-10-06 11:43:27	Event time
Event_Type	FALL / IMPACT / GAS / SOS	Event classification
Severity	LOW / MEDIUM / HIGH /	Alert priority
Accel_Summary	CRITICAL	Motion/impact evidence
Gyro_Summary	8.9 g peak	Fall evidence
Gas_Level	orientation change	Environmental state
Battery	NORMAL / WARNING /	Power status
Signal	CRITICAL	Communication quality
Sequence_No	82%	Duplicate/loss detection
	RSSI/SNR	
	1842	

3.16 Alert Prioritization

Condition	Helmet indicator	Buzzer	Server action
Normal	Green/normal state	Off	No emergency alert
Hazard warning	Yellow blinking	Slow/intermittent	Log warning + notify dashboard
High impact / suspected	Red blinking	Continuous	Immediate emergency alert
fall	Red blinking	Continuous	Immediate emergency alert
Manual SOS	Red or critical state	Continuous	Emergency/evacuation workflow
Critical gas condition	Yellow warning	Short warning tone	Preventive warning; no

			emergency declaration by itself
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4. RESULTS, VALIDATION PLAN AND DISCUSSION

4.1 Prototype Output

The conceptual output consists of three major interfaces: the physical helmet, the control-room dashboard, and the AI hazard-intelligence layer. The helmet visual includes a front headlamp, status LED, buzzer, sensor modules, a rear communication module and battery compartment. The dashboard displays active emergencies, worker identity, zone, sensor values, battery, signal state, and rescue status.



Figure 4.1: Conceptual prototype output showing helmet state, control dashboard, alert scenarios, live map and system flow.

4.2 Validation Strategy

Because the current work is a proposed prototype/research design rather than a completed field deployment, the correct evaluation method is a controlled validation plan. The system should be evaluated component-by-component and then as an integrated system. Metrics should include event-detection sensitivity, false-alarm rate, communication delivery rate, end-to-end alert latency, zone-location accuracy, battery runtime, gas-sensor response, and AI risk-classification performance.

4.3 Test Cases

Test ID	Scenario	Expected result	Metric
T1	Normal walking	No emergency alert	False alarm rate
T2	Controlled safe fall	Fall event detected	Recall/sensitivity

	simulation		
T3	Helmet impact without worker fall	Impact warning according to fusion logic	Precision/false alarm rate
T4	Manual SOS press	Immediate emergency packet + local alarm	Alert latency
T5	Gas warning simulation	Warning state + dashboard event	Detection/response time
T6	Critical gas simulation Weak LoRa signal	Critical alert + local alarm	Detection/response time
T7	Worker moves between zones	Retry/buffer/retransmit strategy	Packet delivery rate
T8	Repeated incidents in one zone	Server updates zone	Zone accuracy
T9	New worker enters high-risk zone	Risk score increases	Risk-model response
T10		Preventive warning generated	Warning latency

4.4 Expected System Behaviour

Parameter	Expected qualitative outcome	Status
Fall detection	Detect strong fall-like motion sequences while minimizing routine-activity alarms	To be experimentally validated
Impact detection	Combine piezo and IMU evidence for strong impact events	To be experimentally validated
Emergency button	Generate immediate worker-initiated alert	Prototype feature
Gas monitoring	Provide prototype warning based on calibrated sensor readings	To be calibrated/validated
Communication	Deliver compact emergency packets over LoRa through site infrastructure	To be field tested
Worker identification	Associate every event with a unique Helmet ID	Architecture defined
Zone tracking	Display worker/event zone using infrastructure mapping	Architecture defined
Risk intelligence	Increase zone risk after repeated incidents and generate preventive warnings	Prototype algorithm

4.5 AI Hazard-Hotspot Scenario

Consider a mine divided into Zone A, Zone B, and Zone C. During a shift, Helmet SSH-014 reports a suspected fall in Zone B. Later, SSH-018 reports an emergency-button activation in the same zone, followed by a high-impact event from SSH-021. The server observes that the incidents are spatially concentrated. The risk engine therefore raises Zone B from NORMAL to HIGH RISK. When SSH-025 approaches Zone B, the server sends a warning such as: “CAUTION: Recent safety incidents reported in this zone. Proceed carefully and follow supervisor instructions.”

If later data shows that the risk was temporary—for example, after maintenance corrected a damaged walkway—the risk score can decay over time. This prevents the system from permanently labeling a repaired area as dangerous.

4.6 Error and False-Alarm Handling

- ☐ Sensor fusion: require agreement between IMU and impact signals before declaring a severe impact when possible.
- ☐ Post-fall inactivity: use a short inactivity window as supporting evidence rather than relying on one acceleration spike.
- ☐ Manual confirmation: when safe, provide a short cancel window for suspected false falls, while manual SOS remains immediately actionable.
- ☐ Communication buffering: store unsent emergency packets locally and retransmit when a link becomes available.
- ☐ Sequence numbers: prevent duplicate server records when a packet is retransmitted.
- ☐ Sensor health monitoring: flag impossible/stuck readings and battery or sensor failures.
- ☐ AI confidence: show the reasons contributing to a zone-risk score rather than displaying an unexplained AI label.

4.7 Discussion

The proposed architecture is technically plausible because its individual building blocks—wearable sensing, IoT helmets, LoRa telemetry, and fall-detection algorithms—have already been investigated in the literature. [1]–[7] The research value lies in their integration around a collective, location-aware risk model. Instead of treating worker events as isolated records, the system treats the mine as a changing risk environment.

The approach also has an important safety distinction: AI should support, not replace, established safety procedures. A high-risk score can prompt caution, inspection, or supervisor review, but it should not autonomously declare a location safe or unsafe without appropriate operational validation. Likewise, sensor data should not be interpreted as medical diagnosis.

5. CONCLUSION AND FUTURE WORK

5.1 Summary of Contributions

- ☐ Designed a connected emergency safety helmet concept for hazardous workers.
- ☐ Defined unique Helmet ID and Worker ID association for multi-worker monitoring.
- ☐ Integrated motion, impact, gas, emergency-button, LED and buzzer functions in one architecture.
- ☐ Proposed LoRa-based event telemetry suitable for compact low-power messages.
- ☐ Defined underground zone-level location mapping instead of assuming GPS availability.
- ☐ Proposed a server dashboard for live worker status and emergency management.
- ☐ Introduced cross-worker AI hazard-hotspot intelligence based on repeated incidents.
- ☐ Defined a risk-score approach that combines severity, frequency, recency, falls, SOS activations and environmental warnings.
- ☐ Defined a controlled validation plan without claiming unmeasured accuracy.

5.2 Conclusion

This research presents an AI-enabled emergency safety helmet designed to provide both immediate and preventive worker safety. At the immediate-response level, the helmet can detect suspected falls and impacts, accept manual emergency requests, monitor environmental conditions, activate local alarms, and transmit compact emergency data to a central server. At the preventive level, the server aggregates incidents from multiple helmets and identifies zones where incidents repeatedly occur.

The central idea is a shared safety memory: an incident involving one worker can contribute to a warning that protects the next worker. This transforms the helmet from a passive protective item into a connected sensing node within an intelligent workplace safety network. The concept is particularly relevant to underground mining, where workers operate in constrained environments and conventional positioning and communication technologies can be challenging.

5.3 Limitations

Limitation	Impact	Mitigation
Prototype gas sensor	May not accurately measure specific mine gases	Use calibrated certified gas sensors
Underground location	GPS unavailable/unreliable	Gateway/anchor/zone positioning
Fall false positives	Routine actions may resemble falls	Sensor fusion + labeled ML dataset
LoRa coverage	Tunnel geometry affects signal	RF survey + repeaters/mesh
Sparse historical data	New zones have weak AI evidence	Use conservative rules and human oversight
Battery constraints	Sensors and radio consume power	Duty cycling + battery optimization
Safety certification	Prototype electronics are not automatically mine-safe	Industrial certification and intrinsic-safety design

5.4 Future Work

- ☐ Build a working PCB and ruggedized enclosure rather than development-board wiring.
- ☐ Replace prototype MQ-135 sensing with calibrated sensors selected for methane, CO, oxygen deficiency and other site-specific hazards.
- ☐ Develop a labeled dataset of worker motion, falls, impacts and normal activities collected with appropriate safety procedures.
- ☐ Train and compare lightweight ML models such as Random Forest, XGBoost, SVM, or tiny neural networks for edge fall classification.
- ☐ Implement gateway-based or anchor-based indoor positioning and evaluate zone accuracy.
- ☐ Develop a secure cloud/server architecture with device authentication, encrypted communication, audit logs and role-based access.
- ☐ Add a temporal AI model that accounts for time-of-day, shift, maintenance status and incident recency.
- ☐ Add explainable risk alerts such as “3 incidents in Zone B in the last 24 hours” rather than an opaque AI score.
- ☐ Conduct controlled mine-like testing followed by authorized field trials.
- ☐ Evaluate battery runtime, thermal behaviour, enclosure protection, communication reliability and electromagnetic compatibility.
- ☐ Investigate helmet removal detection, worker health sensing, voice communication and automated rescue-team routing as optional future modules.

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APPENDIX A – PROPOSED COMPONENTS AND SPECIFICATIONS

Category	Proposed specification
Controller	ESP32 development board
Motion sensor	MPU6050, 3-axis accelerometer + 3-axis gyroscope
Impact sensor	Piezoelectric impact sensor
Gas sensor	MQ-135 prototype; replace with calibrated mine-grade sensors for deployment
Communication	LoRa SX1278 module + antenna
Power	3.7 V 18650 rechargeable Li-ion battery
Charging	TP4056-based prototype charging module
Local alarm	Mini piezo buzzer
Visual indicators	Red/yellow/green status LED concept
Lighting	High-power LED headlamp
User input	Emergency/SOS push button
Server	Central monitoring and event-history service
Dashboard	Live worker status, alerts, map/zone view, history and risk hotspots
AI layer	Rule-based risk score initially; ML model as sufficient data becomes available

APPENDIX B – SAMPLE EVENT LOG

Time	Helmet	Zone	Event	Severity	AI zone state
11:10:03	SSH-014	SEC-B-T04	FALL	High	CAUTION
11:18:42	SSH-018	SEC-B-T04	SOS	HIGH	HIGH RISK
11:43:27	SSH-021	SEC-B-T04	IMPACT	HIGH	HIGH RISK
11:51:05	SSH-025	SEC-B-T04	ZONE ENTRY	HIGH	Preventive warning sent

APPENDIX C – PRESENTATION POINTS

- ☐ Problem: workers face multiple hazards and emergency response can be delayed.
- ☐ Solution: every worker receives an identified smart helmet connected to a central safety server.
- ☐ Immediate safety: fall/impact detection + SOS button + gas warning + local alarm.
- ☐ Visibility: the server knows which helmet/worker generated an event and which zone it came from.
- ☐ AI novelty: incidents from different workers are aggregated to learn risky zones.
- ☐ Prevention: future workers entering a high-risk zone receive a warning before an incident occurs.
- ☐ Key technical stack: ESP32 + MPU6050 + piezo impact sensor + gas sensor + LoRa + server dashboard + AI risk engine.

- Important limitation: the prototype is a research design; certified industrial sensors, location infrastructure and field validation are required for real deployment.
- One-line pitch: “The helmet does not only detect when a worker is in danger—it learns where danger repeatedly happens and warns the next worker before it happens again.”